

Imagine a Particle moving through 2D Space, its position can be tracked in time as coordinates $(x(t), y(t))$. The curve it makes may not be a function, but individually $x(t)$ and $y(t)$ will be.



Certainly y is not a function of x .^{per x of y} But we can look at coordinates $x(t)$ and $y(t)$ individually.

If x and y can be written as functions of a third variable, called a parameter, i.e.

$$x = f(t), \quad y = g(t)$$

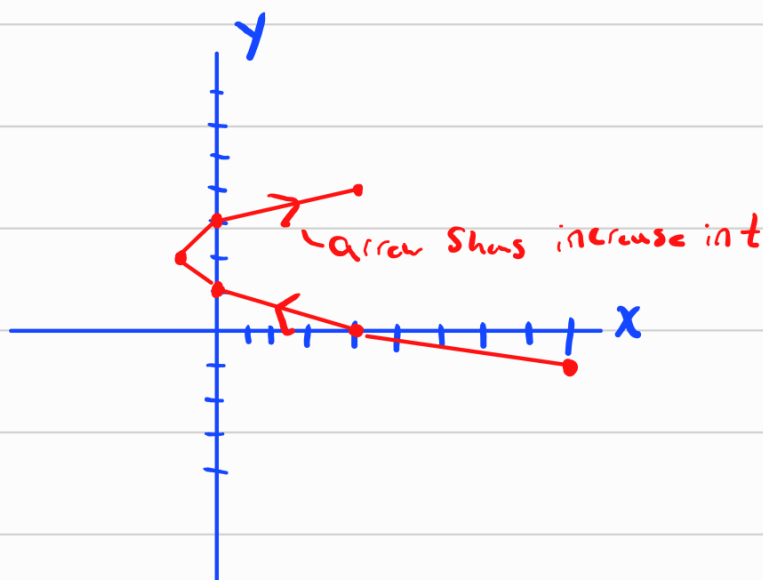
we call this a Parametric equation.

we look at the points as pairs $(x(t), y(t))$

Ex:

$$x = t^2 - 2t, \quad y = t + 1$$

t	x	y
-2	8	-1
-1	3	0
0	0	1
1	-1	2
2	0	3
3	3	4



To get the entire function, add more point.

from $x = t^2 - 2t, \quad y = t + 1$



we can substitute, $t = y - 1$ into x

$$\begin{aligned} x &= (y-1)^2 - 2(y-1) \\ &= y^2 - 4y + 3 \end{aligned}$$

is the curve drawn.

This method, called eliminating the parameter,
is not always straight forward

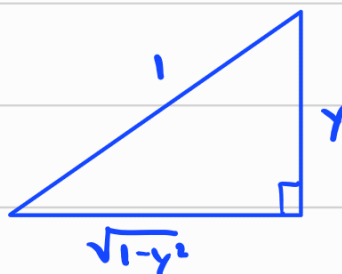
Ex: $x = \cos(t)$, $y = \sin(t)$ $0 \leq t \leq 2\pi$

$$x = \cos(\arcsin y)$$

$$x = \sqrt{1-y^2}$$

$$x^2 = 1 - y^2$$

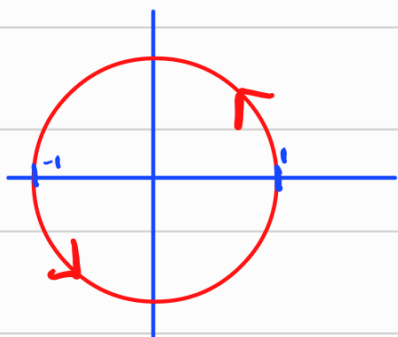
$$x^2 + y^2 = 1$$



or

$$x^2 + y^2 = \sin^2 t + \cos^2 t = 1$$

So, graphically we have



to see the anticlockwise direction, observe $\cos(t)$ and $\sin(t)$ start out positive.

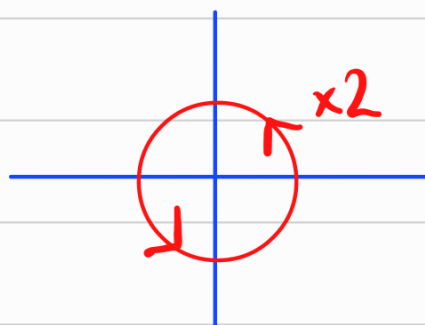
If instead,

$$x = \cos(2t) \quad y = \sin(2t) \quad 0 \leq t \leq 2\pi$$

the shape is the same circle as

$$x^2 + y^2 = \cos^2(2t) + \sin^2(2t) = 1$$

but we go around twice



We always need to consider Particle Motion with Parametric equations. Reason being, two sets of parametric equations can refer to the same curve in different ways.

Ex:

a) $x = t^3$ $y = t$, b) $x = -t^3$ $y = -t$
 c) $x = t^{3/2}$ $y = \sqrt{t}$, d) $x = e^{-3t}$ $y = e^{-t}$

If we eliminate parameters we find

a) $x = y^3$ 
 b) $x = -(-y)^3 = y^3$ 
 c) $x = (y^2)^{3/2} = y^3$ 
 d) $x = e^{-3(-\ln(y))} = e^{\ln(y^3)} = y^3$ 

However,

- a) Particle moves left to right
- b) Particle moves right to left
- c) Particle is only positive, left to right
- d) Particle only positive, right to left and never equals (0,0).

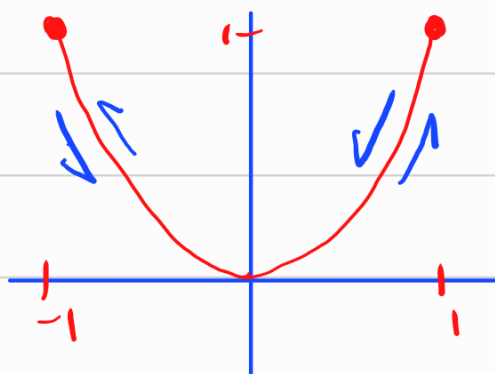
Some parametric functions double back on themselves

Ex: $x = \sin(t)$, $y = \sin^2(t)$

$$x^2 = \sin^2(t) = y$$

So this is a parabola $y = x^2$
but

$$-1 \leq \sin(t) = x \leq 1, \quad 0 \leq y \leq 1$$



Parametric curves give a lot of flexibility as
it is easy to position curves anywhere.

Ex: What is parametric form of a circle
centered at (h,k) with radius r going
anti-clockwise.

The Normal Parametric form of a Circle going Anticlockwise is

$$x = \cos t, \quad y = \sin t, \quad 0 \leq t < 2\pi$$

Since we want radius r , let's take

$$x = r \cos t, \quad y = r \sin t$$

to position at (h, k) , we do

$$x = h + r \cos t, \quad y = k + r \sin t \quad 0 \leq t < 2\pi$$

Most Parametric Equations are hard to visualize even after eliminating the parameter.

Ex:

$$x = \cos(t), \quad y = \sin(2t)$$

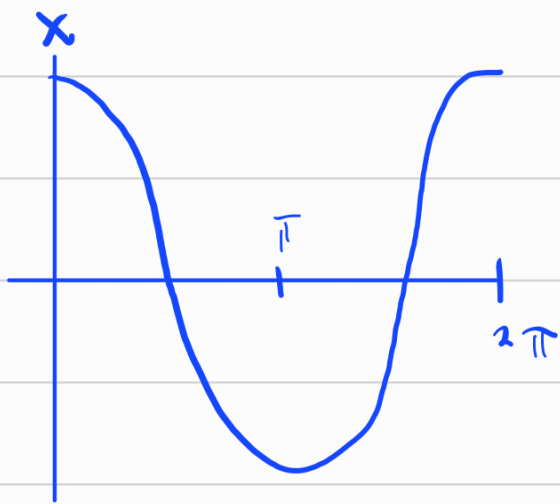
eliminating parameters

$$\underline{y^2 = 4x^2 - 4x^4}$$

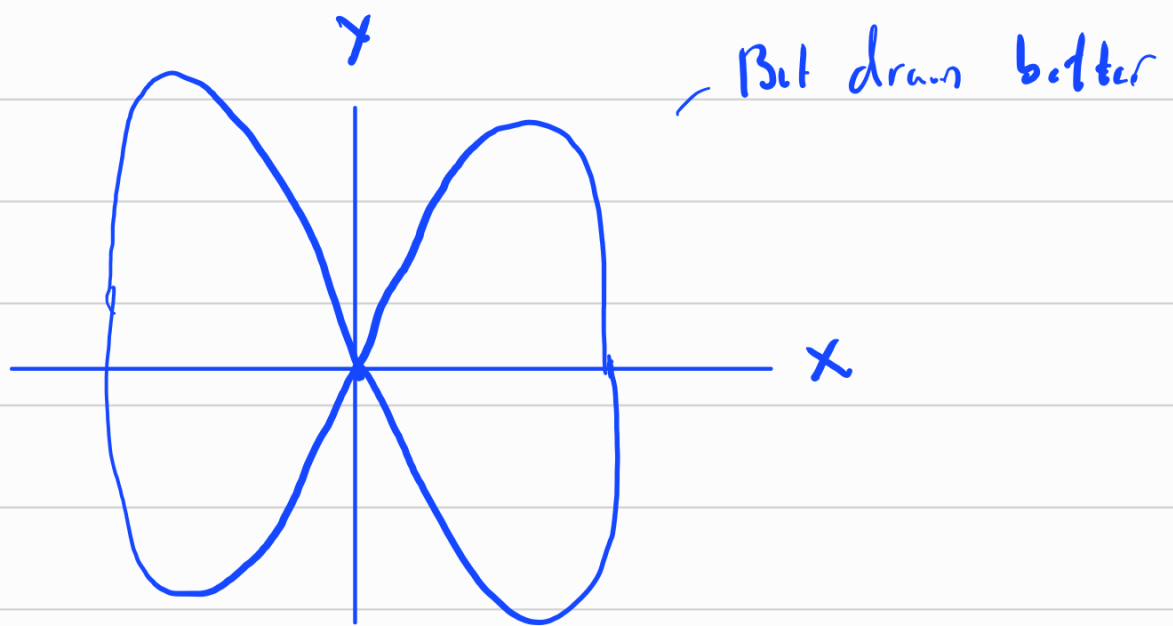
$$\begin{aligned}
y^2 &= \sin^2(2t) \\
&= \frac{1}{2}(1 - \cos(4t)) \\
&= \frac{1}{2}(1 - (2\cos^2(2t) - 1)) \\
&= 1 - \cos^2(2t) \\
&= 1 - [\cos(2t)]^2 \\
&= 1 - [2\cos^2(t) - 1]^2 \\
&= 1 - (4\cos^4(t) - 4\cos^2(t) + 1) \\
&= 2\cos^2(t) - 4\cos^4(t) \\
&= 4x^2 - 4|x^4|
\end{aligned}$$

However, this equation is not helpful.

Let's look at x and y individually



So we expect x to be positive then negative
when y is positive, negative, positive



We can reparameterize things that are already functions. Say, $y = f(x)$ we can parameterize as

$$x = t, \quad y = f(t)$$

likewise, if $x = g(y)$

$$x = g(t), \quad y = t$$

Ex: Parameterize form of

$$y = 3x^2 - \ln(4x+2)$$

$$x = t, \quad y = 3t^2 - \ln(4t+2)$$

Ex:

An ellipse is graphed by
$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$$

write it in parametric form going anti-clockwise

let's start with a circle

$$x = \cos(t) \quad y = \sin(t)$$

The radius changes, so let's put temporary variables

$$x = k \cos t, \quad y = l \sin t$$

in the equation above

$$\left(\frac{k \cos t}{a}\right)^2 + \left(\frac{l \sin t}{b}\right)^2 = 1$$

$$\left(\frac{k}{a}\right)^2 \cos^2 t + \left(\frac{l}{b}\right)^2 \sin^2 t = 1$$

This only makes if $\frac{k}{a} = 1$ and $\frac{l}{b} = 1$

so, $k = a$, $l = b$, and

$$x = a \cos t, \quad y = b \sin t \quad 0 \leq t < 2\pi$$

Intersection vs Collision

An intersection of two parametric equations is a point in which their paths cross for any times.

A collision of two parametric equations is a single time t in which the two equations are intersecting (same point, same time)

Ex:

1. $x = t + 5$ $y = t^2 + 4t + 6$

2. $x = 2t + 4$ $y = 2t + 9$

Take u, v

$$x(u) = u + 5$$

$$y(u) = u^2 + 4u + 6$$

$$x(v) = 2v + 4$$

$$y(v) = v^2 + 4v + 6$$

$$\text{If } x(u) = x(v) \text{ then } u + 5 = 2v + 4 \rightarrow u = 2v - 1$$

$$\text{If } \gamma(u) = \gamma(v)$$

$$u^2 + 4u + 6 = v^2 + 4v + 6$$

$$(2v-1)^2 + 4(2v-1) = v^2 + 4/v$$

$$4v^2 - 4v + 1 + 8v - 4 = v^2 + 4/v$$

$$3v^2 - 3 = 0$$

$$(v+1)(v-1) = 0$$

$$\underline{u = 2v - 1}$$

$$v = 1, \quad u = 1 \quad \text{Collision}$$

$$v = -1, \quad u = -3 \quad \text{Intersection}$$

Ex: The curve

$$x = 2t - t^3 \quad y = t - t^2$$

intersects itself.

$$-t^2 + t - y = 0$$

$$t = \frac{-1 \pm \sqrt{1 - 4y}}{2}$$

,

$$x_1 = 1 - \sqrt{1-4y} - \left(\frac{1 - \sqrt{1-4y}}{2} \right)^3 = 1 + \sqrt{1-4y} - \left(\frac{1 + \sqrt{1-4y}}{2} \right)^3$$

$$-2\sqrt{1-4y} = \frac{1}{8} \left[(1 + \sqrt{1-4y})^3 + (1 - \sqrt{1-4y})^3 \right]$$

$$= \frac{1}{8} \left[(1 + 3\sqrt{1-4y} + 3 - 12y + \sqrt{1-4y}^3) + (1 - 3\sqrt{1-4y} + 3 - 12y - \sqrt{1-4y}^3) \right]$$

$$= \frac{1}{8} \left[-6\sqrt{1-4y} - 2\sqrt{1-4y}^3 \right]$$

$$-2\sqrt{1-4y} = -\frac{3}{4}\sqrt{1-4y} - \frac{1}{4}\sqrt{1-4y}^3$$

$$-2 = -\frac{3}{4} - \frac{1}{4}(1-4y)$$

$$-8 = -3 - 1 + 4y$$

$$-4 = 4y$$

$$\boxed{y = -1}$$

$$\boxed{x = -1}$$

$$\boxed{t = \frac{1 \pm \sqrt{1-4y}}{2} = \frac{1 \pm \sqrt{5}}{2}}$$

$$x = 2t - t^3 = 1 + \sqrt{5} - \frac{1}{8}(1 + \sqrt{5})^3$$

$$= 1 + \sqrt{5} - \frac{1}{8}(1 + 3\sqrt{5} + 15 + 5\sqrt{5})$$

$$= -1$$